

**BTP Presentation
on**



Mixture of Experts for Weather Prediction

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Roll Number
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Introduction



- Deep learning model for hourly weather forecasting
- Predicts 6 parameters together
 - Temperature, Humidity, Pressure, Wind Speed, Direction, Rainfall
- Uses past 48 hours data
- Needed for:
 - Agriculture
 - Renewable energy
 - Aviation
- Forecast errors in these domains carry direct economic and safety consequences

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Literature Review



Reference (Author(s), Volume number, Year)	Schematic / Diagram / Flowchart / Dataset / etc.	Methodology	Salient Features
FourCastNet (Pathak et al., 2022)	Global ERA5 grid (0.25°); multi-variable forecast	Adaptive Fourier Neural Operators	Not designed for single-site surface forecasting; no regime-aware gating
Pangu-Weather (Bi et al., 2023)	3D pressure-level ERA5 dataset	3D Earth-Specific Transformers with separate spatial and temporal attention heads	Fixed architecture; regime-invariant; not applicable to single-location hourly surface prediction
Lam et al. Science, vol. 382, 2023	Icosahedral multi-mesh graph (ERA5)	Graph Neural Networks on a multi-scale icosahedral mesh	flexible spatial resolution; requires global gridded input
EWMoE (Chen et al., 2024)	MoE in transformer weather model, ERA5 evaluation benchmark	Sparse MoE layers inserted inside a transformer	Competitive with Pangu-Weather on medium-range forecasts at lower compute

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Problem Definition



- Existing DL weather models are primarily designed for **global, medium-range forecasting** and are not optimized for **single-site, high-frequency surface prediction**
- Identified gaps from literature:
 - Fixed model behavior across conditions
 - Implicit modeling of inter-variable relationships without leveraging known physical dependencies
- These limitations impact performance, especially for **state-dependent variables**.
- Existing MoE approaches do not clearly use atmospheric conditions to guide expert selection.
- **NWP** systems are computationally expensive and have limited ability for rapid local updates while the **Global DL** models are resource-intensive and not specifically calibrated for surface-level, single-location forecasting

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Methodology



- Input: **48-hour historical data**
- Output: **Next-hour prediction (6 variables)**
- **NASA POWER**: 10 years (2015-2025), hourly, single location at 21.5°N 70.9°E (Diu), 6 variables, train/val/test split 2015-2022 / 2023 / 2024–2025
- **Diu Ground Station**: IIITV-ICD campus, Oct 2024 - Apr 2026, 2-minute resolution, 9 variables including 3 air quality variables, 80/10/10 split

Core Components:

- Feature Engineering
- Temporal Attention
- Cross-Parameter Attention
- Mixture of Experts (MoE)
- Physics Constraints

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Methodology Continued



- **Feature Engineering**
 - Lag, moving avg, cyclic features
- **Temporal Attention**
 - Short-term (24h) + Long-term (48h)
- **MoE Model**
 - 4 expert networks
 - Gating based on weather regime
- **Physics Constraints**
 - Prevent invalid predictions

Tools:

- Python, PyTorch, Google Colab

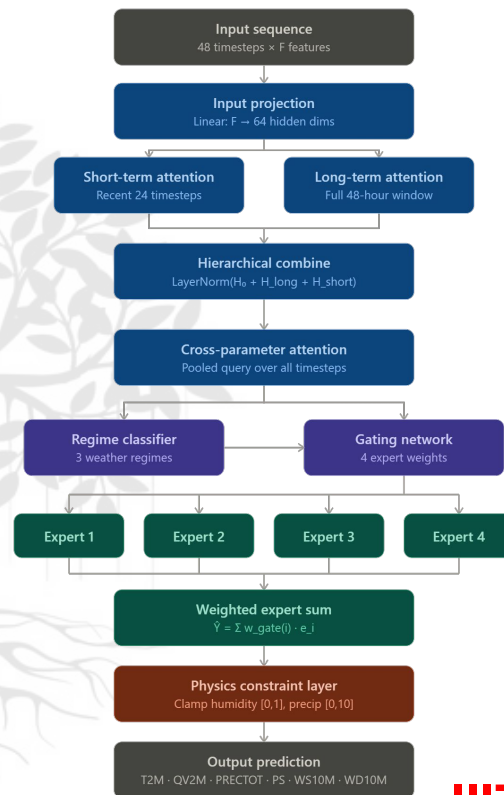


Fig 2: Flow Diagram of model

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Results



- We used 2 datasets NASA POWER dataset (10 Years) and the Diu Ground Station dataset (IIITV-ICD)
- After training Model gets High accuracy for Temperature ($R^2 \approx 0.93$ – 0.96) and Humidity ($R^2 \approx 0.97$), While it performs Moderately on Wind Speed, Pressure
- And it faced Challenges for the variables: Precipitation (skewed data) and Wind Direction (circular nature)
- Similarly for the diu dataset its overall performance was good for variables other than the air quality

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Results



Parameter	RMSE	MAE	R ²
Temperature (°C)	0.669	0.529	0.936
Specific Humidity (g/kg)	0.720	0.577	0.972
Precipitation (mm/hr)	7.704	0.937	0.531
Surface Pressure (kPa)	0.198	0.162	0.844
Wind Speed (m/s)	1.008	0.788	0.784
Wind Direction (°)	37.20	14.81	0.890

Table: Model Performance for NASA Dataset

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Results

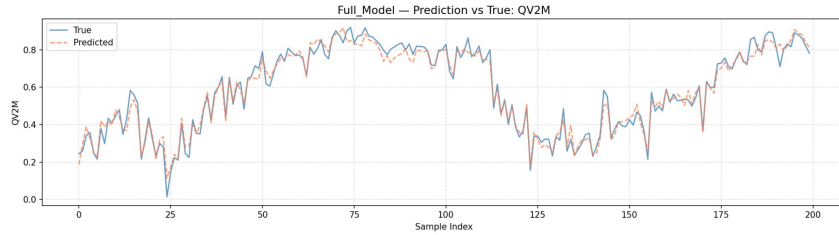


Fig : Prediction vs True: QV2M (Humidity)

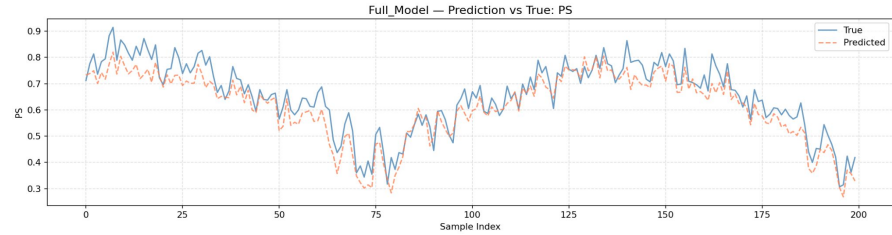


Fig : Prediction vs True: PS (Pressure)

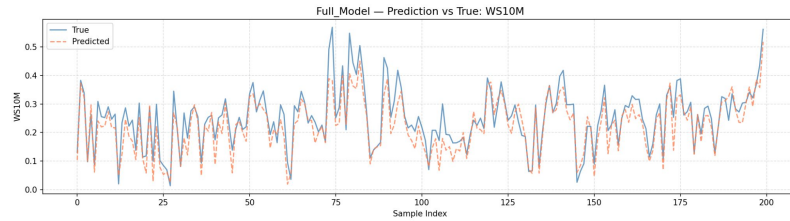


Fig : Prediction vs True: WS10M (wind speed)

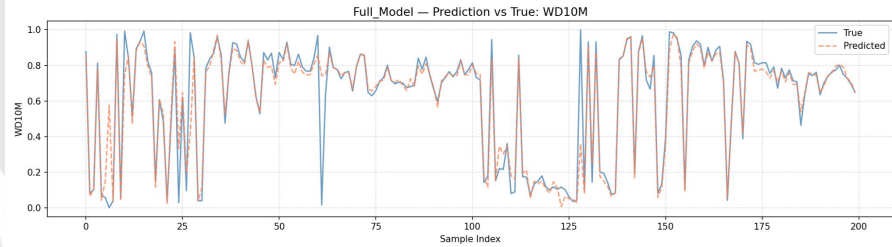


Fig : Prediction vs True: WD10M (Wind Direction)

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Results

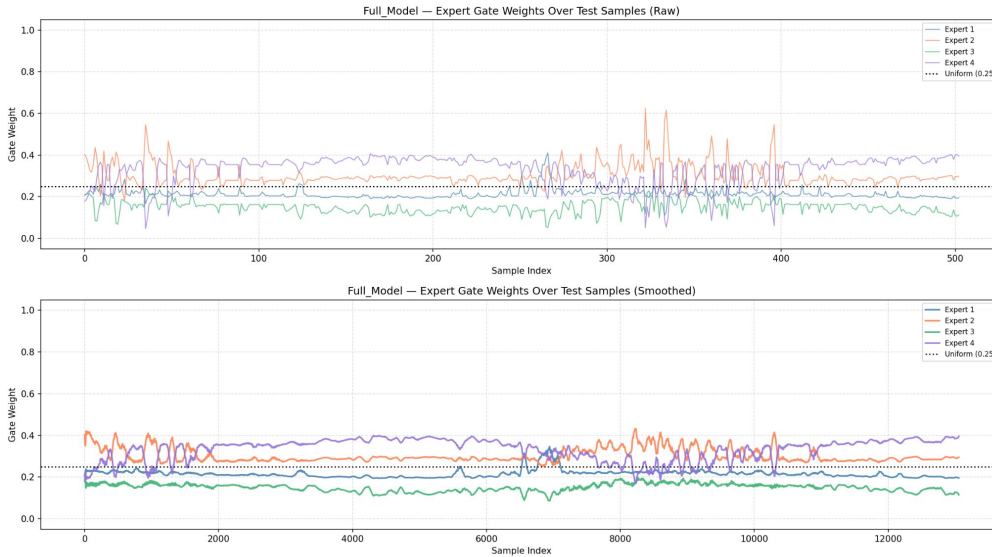


Fig: Expert gate weight over Test samples

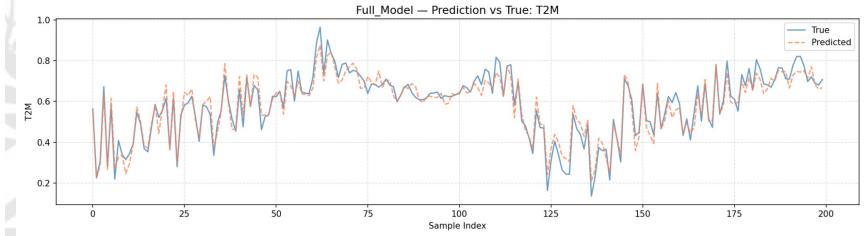


Fig : Prediction vs True: T2M (Temperature)

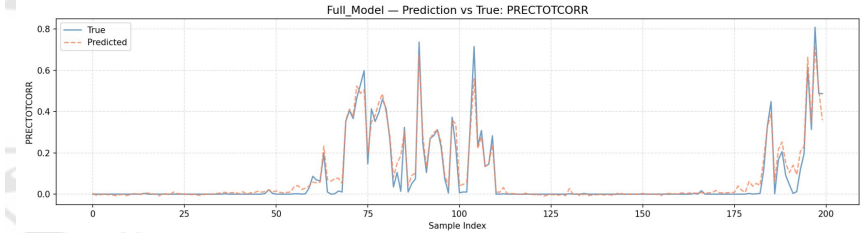


Fig : Prediction vs True: PRECTOTCORR (precipitation)

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Results



Parameter	RMSE	MAE	R ²
Temperature (°C)	0.791	0.596	0.961
Humidity (%)	4.468	3.636	0.913
Wind Speed (m/s)	0.693	0.528	0.542
Wind Direction (°)	57.92	37.41	0.327
Solar Radiation (W/m ²)	34.22	18.44	0.982
Rainfall (mm)	0.055	0.025	0.653
PM2.5 (µg/m ³)	2.506	1.993	0.082
PM10 (µg/m ³)	3.568	2.814	0.071
AQI	5.316	4.177	0.075

Table: Model Performance for Diu Ground Station Dataset

Conclusion



- Built efficient **weather prediction model**
- Handles:
 - a. Multi-variable prediction
 - b. Temporal patterns
 - c. Weather regimes
- **Key Findings:**
 - a. MoE improves complex variables
 - b. Attention improves dependencies
 - c. Model is lightweight & fast
- **Outcome:**
Successfully solves identified gaps

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Future Work



- Multi-location prediction
- Multi-step forecasting
- Better handling of extreme events

Improvements:

- Better loss for rainfall
- Wind direction optimization

Real-world applications

- Precision agriculture - automated frost alerts, irrigation scheduling
- Solar energy forecasting - irradiance prediction for grid balancing
- Aviation and marine safety - terminal area wind and pressure advisories

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References



- [1] J. Pathak et al., "FourCastNet: A global data-driven high-resolution weather model using adaptive Fourier neural operators," arXiv:2202.11214, 2022.
- [2] K. Bi et al., "Pangu-Weather: A 3D high-resolution model for fast and accurate global weather forecast," Nature, vol. 623, pp. 688–694, 2023.
- [3] R. Lam et al., "Learning skillful medium-range global weather forecasting," Science, vol. 382, no. 6677, pp. 1416–1421, 2023.
- [4] I. Price et al., "GenCast: Diffusion-based ensemble forecasting for medium-range weather," arXiv:2312.15796, 2023.
- [5] S. Hochreiter and J. Schmidhuber, "Long short-term memory," Neural Computation, vol. 9, no. 8, pp. 1735–1780, 1997.
- [6] X. Chen et al., "Enhanced weather forecasting with mixture-of-experts neural networks," arXiv:2405.06004, 2024.